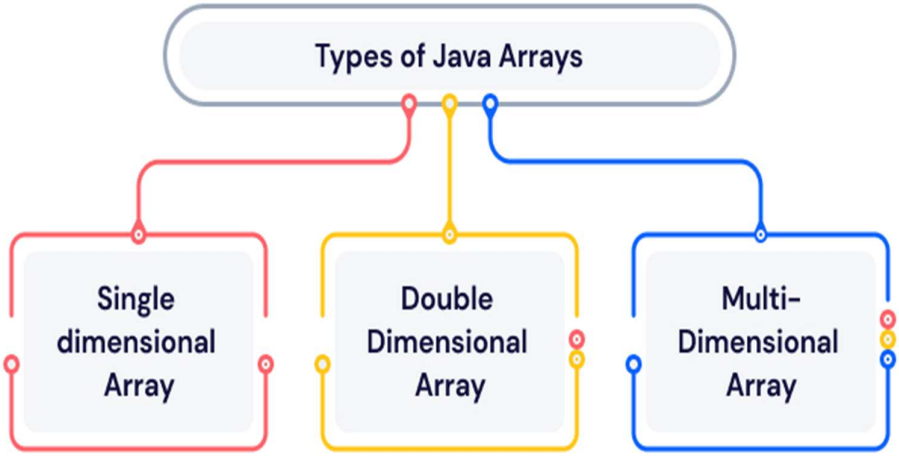
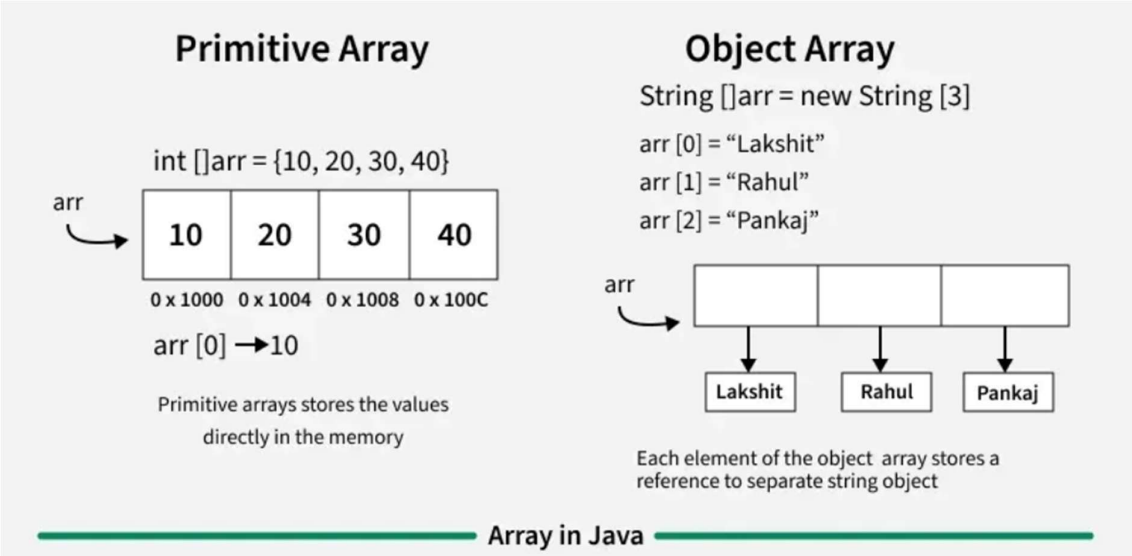
TYPES OF ARRAYS

- **There are three types of Array :-**





2 Dimensional Array :-

- M A 2D array is also known as a matrix (a table of rows and columns).
- **Syntax :-**

```
Datatype arrayname [][] = {{value 1, value 2},
                           {value 3, value 4}};

Datatype arrayname[][] = new datatype [Size of row][Size of
                                                    column];
```

- **Example :-**

```
Int arr [][] = {{1,2},{3,4}};
```

Multi Dimensional Array :-

- In Multi dimensional array in any array with more than one dimension.

- **Syntax:-**

```
Datatype array name [][][]= {{{value1, value2},
                              {value3, value4}},
                             {{value5, value6},
                              {value7, value8}}};
```

- **Example :-**

```
int arr [][][]= {{{10, 20},
                  {30, 40}},
                 {{50, 60},
                  {70, 80}}};
```

Implementation

Input :-

```java
package basicconcepts;

public class array2d {
    public static void main(String[] args) {
        // Two Dimensional Array

        int arr[][]= { {10,20},
                       {30, 40}
        };

        for(int r=0;r<arr.length;r++)
        {
            for(int c=0;c<arr.length;c++)
            {
                System.out.print(arr[r][c]+" ");
            }
            System.out.println();
        }
    }
}
```

Output :-

```
10 20
30 40
```

Input :-

```java
package basicconcepts;

public class array2dnew {
    public static void main(String[] args) {

        long num[][]= new long[2][2];
        num[0][0]=56;
        num[0][1]=42;
        num[1][0]=36;
        num[1][1]=48;

        for(int r=0;r<num.length;r++)
        {
            for(int c=0;c<num.length;c++)
            {
                System.out.print(num[r][c]+" ");
            }
            System.out.println();
        }
    }
}
```

Output :-

```
56 42
36 48
```

Task 1

Input :-

```java
package assignment;

public class taskarray2d {
    public static void main(String[] args) {

        String arr[][] = {
            {"Id", "Name", "Surname"},
            {"1", "Nikhil", "Shinde"},
            {"2", "Omkar", "Pawar"}
        };

        for(int i = 0; i < arr.length; i++)
        {
            for(int j = 0; j < arr.length; j++)
            {
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output :-

```
Id Name Surname
1 Nikhil Shinde
2 Omkar Pawar
```

Task 2

Input :-

```java
package assignment;

public class taskmutlidimensional {
    // Multidimensional Array
    public static void main(String[] args) {

        int a [][][] = { { {1,2},
                           {3,4}},
                         { {5,6},
                           {7,8}} };

        for(int i=0;i<a.length;i++)
        {
            for(int j=0;j<a.length;j++)
            {
                for(int k=0;k<a.length;k++)
                {
                    System.out.print(a[i][j][k] +" ");
                }
                System.out.println();
            }
            System.out.println();
        }
    }
}
```

Output :-

```
1 2
3 4

5 6
7 8
```